

DBMS — Lecture 5

Data Definition Language (DDL) & Data Manipulation Language (DML)

Learning Objectives

Understand DDL and DML.

Identify common SQL commands.

Write basic SQL statements.

SQL

SQL (Structured Query Language) is widely used to communicate with relational databases.

DDL — Data Definition Language

DDL defines and modifies the structure of database objects.

CREATE: `CREATE TABLE Student (RollNo INT, Name VARCHAR(50), Branch VARCHAR(30));`

ALTER: `ALTER TABLE Student ADD Marks INT;`

DROP: `DROP TABLE Student;`

TRUNCATE: `TRUNCATE TABLE Student;`

DML — Data Manipulation Language

DML is used to insert, modify, delete, and retrieve data.

INSERT: `INSERT INTO Student VALUES (101, 'Raju', 'CSE');`

UPDATE: `UPDATE Student SET Branch='IT' WHERE RollNo=101;`

DELETE: `DELETE FROM Student WHERE RollNo=101;`

SELECT: `SELECT * FROM Student;`

DDL vs DML

DDL	DML
Manages database structure	Manages stored data
CREATE, ALTER, DROP, TRUNCATE	INSERT, UPDATE, DELETE, SELECT*
Focus: schema/objects	Focus: records/data

Important Note

*In many introductory courses SELECT is grouped with DML; more formally, SELECT is often treated as DQL (Data Query Language).

Key Takeaway

DDL manages database structure, while DML manages the data stored inside that structure.